

Dexter Fernandes

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Computer Vision Data Scientist

Computer Vision Data Scientist with experience building end-to-end low-latency vision solutions on large real-world datasets for British Petroleum. Strong background in designing, training, and deploying production CV models including YOLOv5/v8, Faster R-CNN, EfficientDet, SSD MobileNet V2, and ByteTrack using Python, PyTorch, TensorFlow, OpenCV, and NumPy. Worked with multinational stakeholders, from project managers to drill rig operators, translating business problems into data science solutions and communicating findings to technical and non-technical audiences. Delivered production systems with gains in false-positive reduction, rollout speed, and reliability, with additional experience in RAG and LLM workflows.

WORK EXPERIENCE

Esbaa • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Owned the end-to-end computer vision pipeline for real-time perception systems in harsh industrial environments, from data curation and model training through deployment and on-site validation, improving F1 by 15% since project takeover while maintaining deterministic 24/7 operation, using PyTorch.
- Built multimodal data ingestion, cleansing, training, and validation pipelines using Python, Pandas, Apache Airflow, MLflow, DVC, Optuna, and CVAT, cutting model iteration time by 40% and reducing rollout time from 3 days to under 24 hours, handling upto 3.6 million records.
- Applied statistical and experimental methods including Bayesian optimisation, ablation studies, hypothesis testing, and confidence intervals to support model selection, threshold tuning, and failure analysis in production.
- Worked with multinational stakeholders, from project managers to drill rig operators, translating operational problems into deployable data science solutions and communicating findings to both technical and non-technical stakeholders.
- Built synthetic image generation and augmentation procedures with OpenCV and SciPy to improve model robustness under harsh industrial conditions, reducing false positives and supporting long-term operational reliability.
- Applied classical image-processing techniques to detect mud and sand splashes on camera lenses, reducing false positives by 13% and improving system reliability in harsh industrial conditions.

LivNSense Technologies • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI

- Led a 4-member team to design, train, and deploy production object detection and tracking models for industrial safety applications, reducing false positives by 33% and achieving sub-50 ms inference latency on edge devices.
- Built low-latency perception pipelines using OpenVINO, and C++17, increasing throughput from 5 FPS to 15 FPS and delivering deterministic real-time inference on Intel VPU hardware under tight power and memory constraints.
- Optimised model architectures using quantisation-aware training, structured pruning, layer fusion, FP16/INT8 deployment, and targeted profiling, reducing model size by 50% while meeting accuracy, latency, and memory constraints.
- Built automated labelling, dataset versioning, and performance-tracking workflows, repurposing existing detection models for annotation and cutting manual labelling effort by 80% while accelerating iteration cycles on large real-world datasets.
- Collaborated with hardware, software, and QA teams to integrate CV models into deployed systems, and presented performance findings to leadership and clients to support rollout, optimisation, and reliability decisions.
- Evaluated edge versus cloud inference trade-offs to minimise bandwidth, latency, and operational cost while maintaining safety-critical performance, supporting production cloud-hosted ML workloads on Microsoft Azure.

PROJECTS

Modern Edge CV Benchmarking

Mar 2026

- Benchmarked modern computer vision models including YOLOv8, YOLOv10, and RT-DETR on a shared validation subset, comparing mAP, inference time, and model size to understand trade-offs for low-latency deployment.
- Exported models to ONNX and evaluated edge deployment paths using OpenVINO, strengthening practical understanding of optimisation and model selection for production-style CV systems.
- Extended the work by studying recent edge and foundation-model approaches including Grounding DINO, MobileSAM, Depth Anything v2, and EfficientViT to stay current with state-of-the-art computer vision methods.

End-to-End Damage Severity & Claim Assessment

Nov 2025

- Developed an end-to-end automotive claims assessment pipeline combining computer vision-based damage detection with a trained severity-classification model, achieving accurate triage for real-world vehicle damage scenarios.
- Built a lightweight repair-insight engine that transforms model outputs into insurer-friendly explanations, using structured reasoning + LLM-generated narratives to deliver clear, interpretable claim summaries.
- Designed the entire workflow (data preprocessing, model training, evaluation, and a user-facing inference module) demonstrating practical understanding of insurance claims processes.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Dissertation: Radiological SLAM with LiDAR Odometry

- Radiological SLAM with LiDAR Odometry. Designed and implemented a real-time SLAM system using RTAB-Map on the Clearpath Husky platform, integrating LiDAR and radiological data for localisation and mapping in inspection scenarios.
- Evaluated RTAB-Map, LIO-SAM, and Cartographer across comparable scenarios to assess mapping and localisation behaviour in simulation before real-world deployment.
- Validated sensor placement for coverage and calibrated LiDAR extrinsics to the robot base, including the Velodyne VLP-16 and Intel RealSense D435i.
- Implemented a ROS2 (rclcpp) point cloud pipeline in C++17 using PCL for filtering, segmentation, and registration.
- Developed and maintained ROS2 nodes using tf2 and message synchronisation, and debugged tf trees and frame connectivity.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

- Built a strong foundation in signals, systems, embedded hardware, and control, which later supported work in real-time AI, perception systems, and robotics-adjacent engineering.

CERTIFICATIONS

Building RAG Agents with LLMs

Mar 2026 - Apr 2026

Nvidia

Deep Learning Specialisation

Jul 2019

deeplearning.ai

AWARDS & SCHOLARSHIPS

Best Speaker award

Jan 2020

ABC Toastmasters club

SKILLS

- **Deep Learning and Optimisation:** Bayesian Optimisation, Convolutional Neural Networks, Data Augmentation, Layer fusion, MultiRocket, Object Detection & Tracking, ONNX, OpenVINO, Optuna, Post-training Quantisation, PyTorch, Quantisation-Aware Training, Scikit-Learn, Structured pruning, TensorFlow, TensorRT, YOLO
- **Computer Vision Models:** ByteTrack, Depth Anything v2, EfficientDet, EfficientViT, Faster R-CNN, Grounding DINO, MobileSAM, RT-DETR, SSD MobileNet V2, YOLOv10, YOLOv5, YOLOv8
- **Cloud, Edge & MLOps:** Apache Airflow, Apache Kafka, API design, AWS, CI/CD, Docker, DVC (Data Version Control), Intel VPUs, Microsoft Azure, MLFlow
- **Programming & Tooling:** Bash, C++17, CMake, colcon, Django, FastAPI, Git, Intel VTune, Linux, NumPy, Pandas, PG vector, PostgreSQL, Python, Qt, SciPy
- **Professional Skills:** Agile framework, Experiment-driven development, Stakeholder communication, Technical leadership
- **Large Language Models:** Citation-grounded response generation, Error analysis, Experimentation and eval, Information extraction, Model fine-tuning, Retrieval-augmented generation (RAG), Search and Retrieval, Workflow automation
- **AI-Assisted Development:** ChatGPT Codex, Claude Code, Pompt engineering